

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: G Vishnu Madhav Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I G Vishnu Madhav (192425192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Vishnu Madhav

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ASSIGNMENT – MLE AND WEIGHT UPDATE

QUESTION 1: Estimating Probability of Heads Using MLE

Question:

A coin is tossed 10 times: H H T H H H T H H T.

Estimate the probability of Heads using MLE.

Answer

Given:

$$n = 10$$

$$x = 7 \text{ Heads}$$

Theory:

MLE estimates the value of p that makes the observed data most likely.

Likelihood Function:

$$L(p) = p^x (1-p)^{(n-x)}$$

Substitute values:

$$L(p) = p^7 (1-p)^3$$

og Likelihood:

$$\log L(p) = 7 \log(p) + 3 \log(1-p)$$

Differentiate and equate to zero:

$$7/p - 3/(1-p) = 0$$

$$7(1-p) = 3p$$

$$7 - 7p = 3p$$

$$7 = 10p$$

$$p = 7/10 = 0.7$$

Result:

MLE of probability of Heads = 0.7 (70%).

QUESTION 2: Probability of Passing an Exam Using MLE

Given:

$$n = 50$$

$$x = 42$$

Likelihood:

$$L(p) = p^{42}(1-p)^8$$

MLE:

$$p = x/n$$

$$p = 42/50 = 0.84$$

Result:

Estimated probability of passing = 0.84 (84%)

QUESTION 3: Probability of Defective Bulbs Using MLE

Given:

$$n = 100$$

$$x = 8$$

Likelihood:

$$L(p) = p^8(1-p)^{92}$$

MLE:

$$p = x/n$$

$$p = 8/100 = 0.08$$

Result:

Estimated probability of a defective bulb = 0.08 (8%).

QUESTION 4: Probability of Rolling a 6 Using MLE

Given:

$$n = 60$$

$$x = 18$$

Likelihood:

$$L(p) = p^{18}(1-p)^{42}$$

MLE:

$$p = x/n$$

$$p = 18/60 = 0.30$$

Result:

Estimated probability of rolling a 6 = 0.30 (30%).

QUESTION 5: Neural Network Weight Update

Given:

$$\text{Input } x = 4$$

$$\text{Weight } w = 2$$

$$\text{Actual Output} = 10$$

$$\text{Gradient} = 4$$

$$\text{Learning Rate} = 0.01$$

Prediction:

$$y = w \times x$$

$$y = 2 \times 4 = 8$$

Error:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

$$\text{Error} = 10 - 8 = 2$$

Weight Update:

$$w(\text{new}) = w - (\text{Learning Rate} \times \text{Gradient})$$

$$w(\text{new}) = 2 - (0.01 \times 4)$$

$$w(\text{new}) = 2 - 0.04$$

$$w(\text{new}) = 1.96$$

Result:

Predicted Output = 8

Error = 2

Updated Weight = 1.96